

COM 3240

REINFORCEMENT LEARNING

IMMEDIATE REWARDS

SLOT MACHINES: ONE-ARM BANDITS



DALL·E

IMMEDIATE REWARDS

BANDITS

$$E[R] = \sum_i r_i P(R = r_i) \quad \text{Unknowns: } r_i, P(R = r_i)$$

$$Q = \frac{1}{n} \sum_{i=1}^n r_i \quad \text{Stationary}$$

$$\text{Error}_b = \frac{1}{2} \sum_i^n (r_i - Q)^2 \quad \text{Minimise}$$

$$\text{Error}_o = \frac{1}{2} (r - Q)^2 \quad \text{non-Stationary}$$



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OPTIMISATION

GRADIENT DESCENT

Desirable: A system (an agent) that observes rewards and estimates expectations

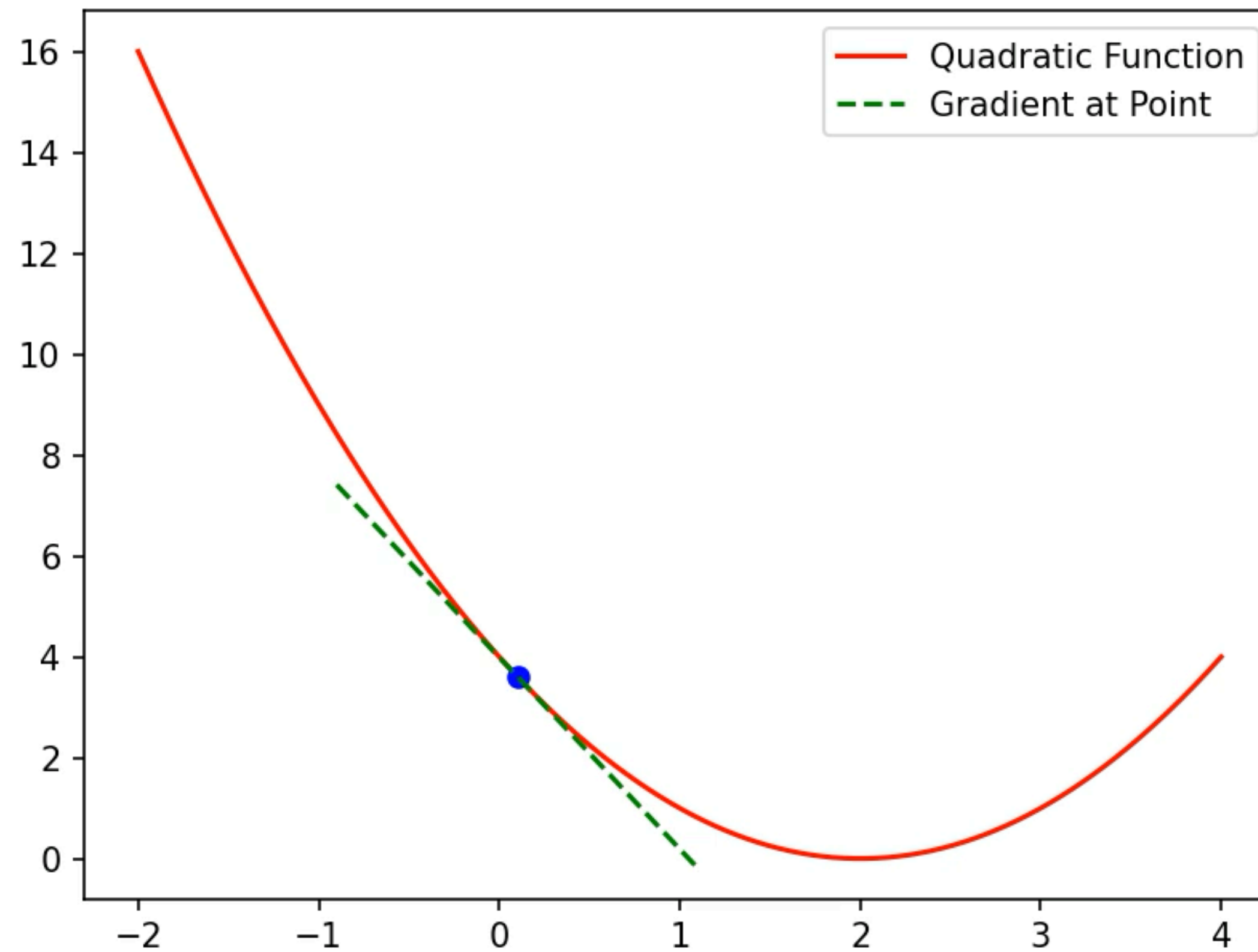
The system has parameters that we need to select appropriately (optimise)

$f(x_1, x_2, \dots, x_n)$
parameters

f System's performance: maximise
System's error : minimise

OPTIMISATION

GRADIENT DESCENT



$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\Delta \mathbf{x} = -\eta \nabla f(\mathbf{x})$$

$$\Delta \mathbf{x} = \mathbf{x}_{i+1} - \mathbf{x}_i$$

POLICIES

THE EXPLORATION EXPLOITATION DILEMMA



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POLICIES

THE EXPLORATION EXPLOITATION DILEMMA

Greedy $a_t = \operatorname{argmax}_a Q_t(a)$

Optimistic Greedy: initialise Q-values unrealistically high

Epsilon-Greedy : explore with probability epsilon, greedy otherwise

Softmax: $P(a) = \frac{e^{Q_t(a)/\tau}}{\sum_b e^{Q_t(b)/\tau}}$

IMMEDIATE REWARDS

BANDITS - STATIONARY ENVIRONMENT

- Initialise Q-values, and count N
- For each time step
 - Select an action using policy
 - Take action and observe reward
 - Increase the count N

$$\mathbf{Error}_b = \frac{1}{2} \sum_i^n (r_i - Q)^2$$

$$\Delta Q = -\eta \frac{\partial \mathbf{Error}_b}{\partial Q} = \eta \sum_{i=1}^n (r_i - Q)$$

- Update estimates $Q(A_{t+1}) = Q(A_t) + (r_t - Q(A_t))/N(A_t)$

$$Q = \frac{1}{n} \sum_{i=1}^n r_i$$

IMMEDIATE REWARDS

BANDITS - NON STATIONARY ENVIRONMENT

- Initialise Q-values, and count N
- For each time step
 - Select an action using policy
 - Take action and observe reward
 - Increase the count N
 - Update estimates $Q(A_{t+1}) = Q(A_t) + \eta(r_t - Q(A_t))$

$$\text{Error}_o = \frac{1}{2}(r - Q)^2$$

FUTURE REWARDS

CHESS, BACKGAMMON, MAZES



Stochastic

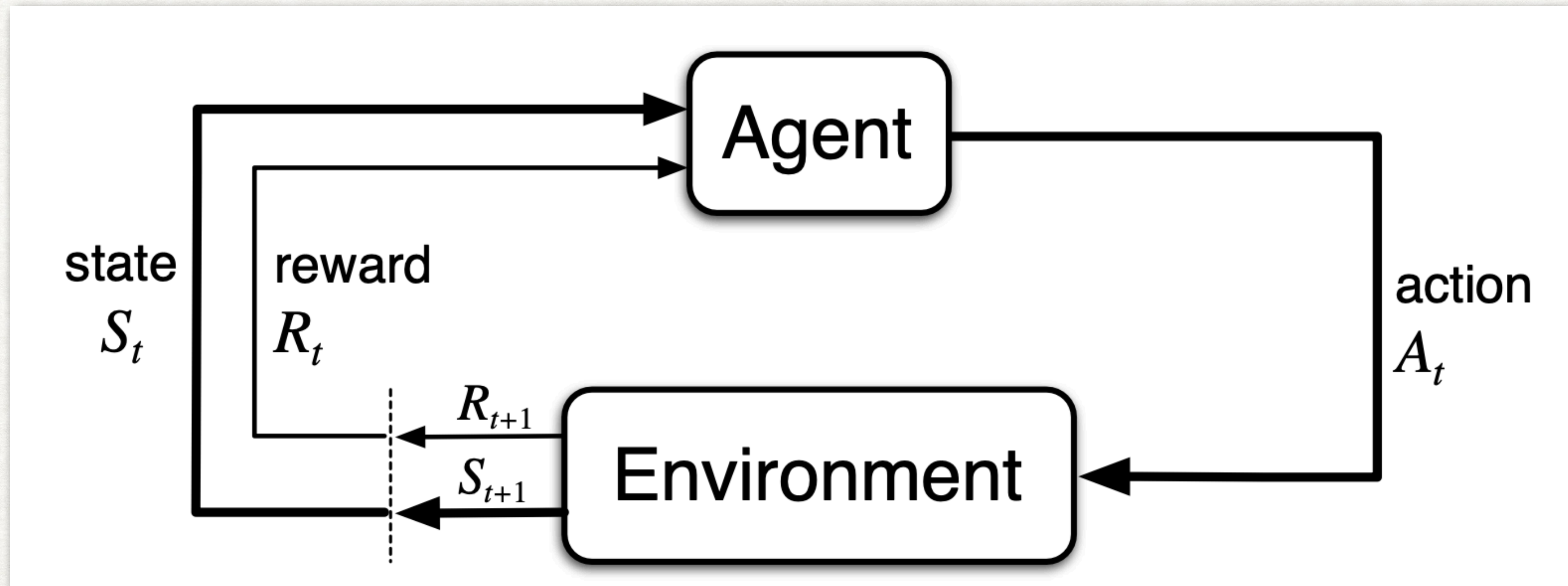


Deterministic

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FUTURE REWARDS

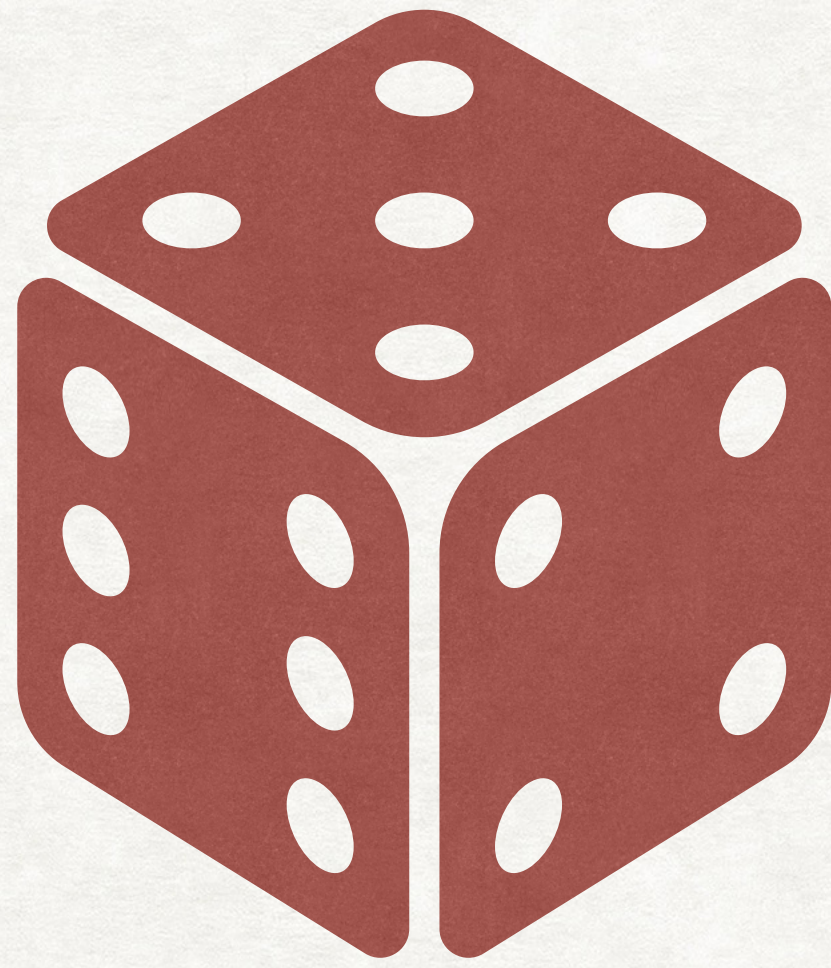
MARKOV DECISION PROCESS



The agent-environment interaction from Sutton & Barto, 2020 - with permission

FUTURE REWARDS

MARKOVIAN PROPERTY



Memoryless

Like a dice; each roll does not depend on the previous ones.

Not true in real life and in many practical scenarios, but there is a remedy.

FUTURE REWARDS

MARKOVIAN PROPERTY

- **Similar to the approach we took for bandits we will see:**
 - **Scenarios of known environment, i.e. known transition probabilities and rewards.**
 - **Equations allowing to estimate total expected rewards.**
 - **Next week we will approximate these equations via an algorithm, so we can use them in “unknown” environments.**

FUTURE REWARDS

MARKOV DECISION PROCESS



2 dials $\{0,1\}$

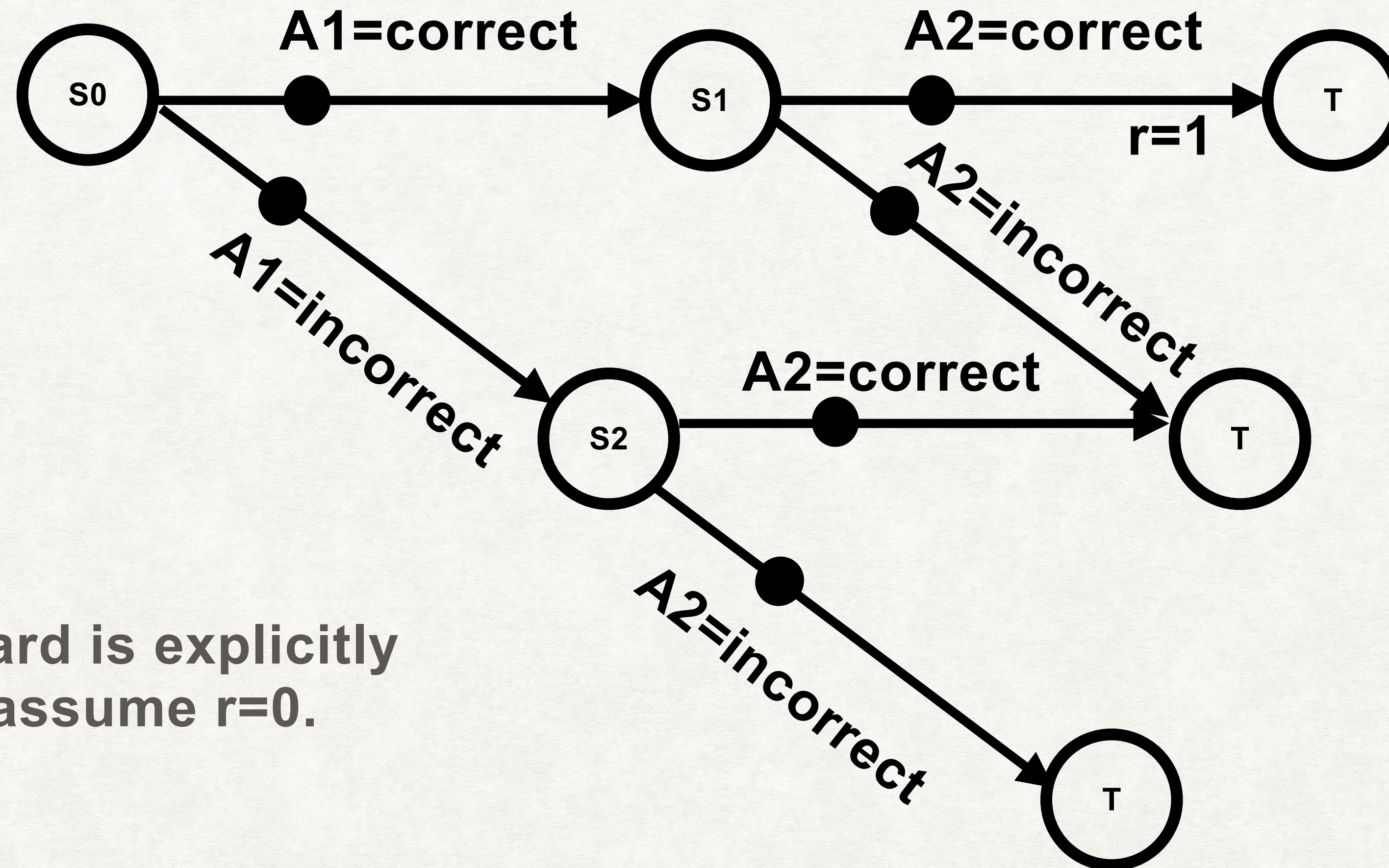
2 digits code

Example of code $[1,0]$

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FUTURE REWARDS

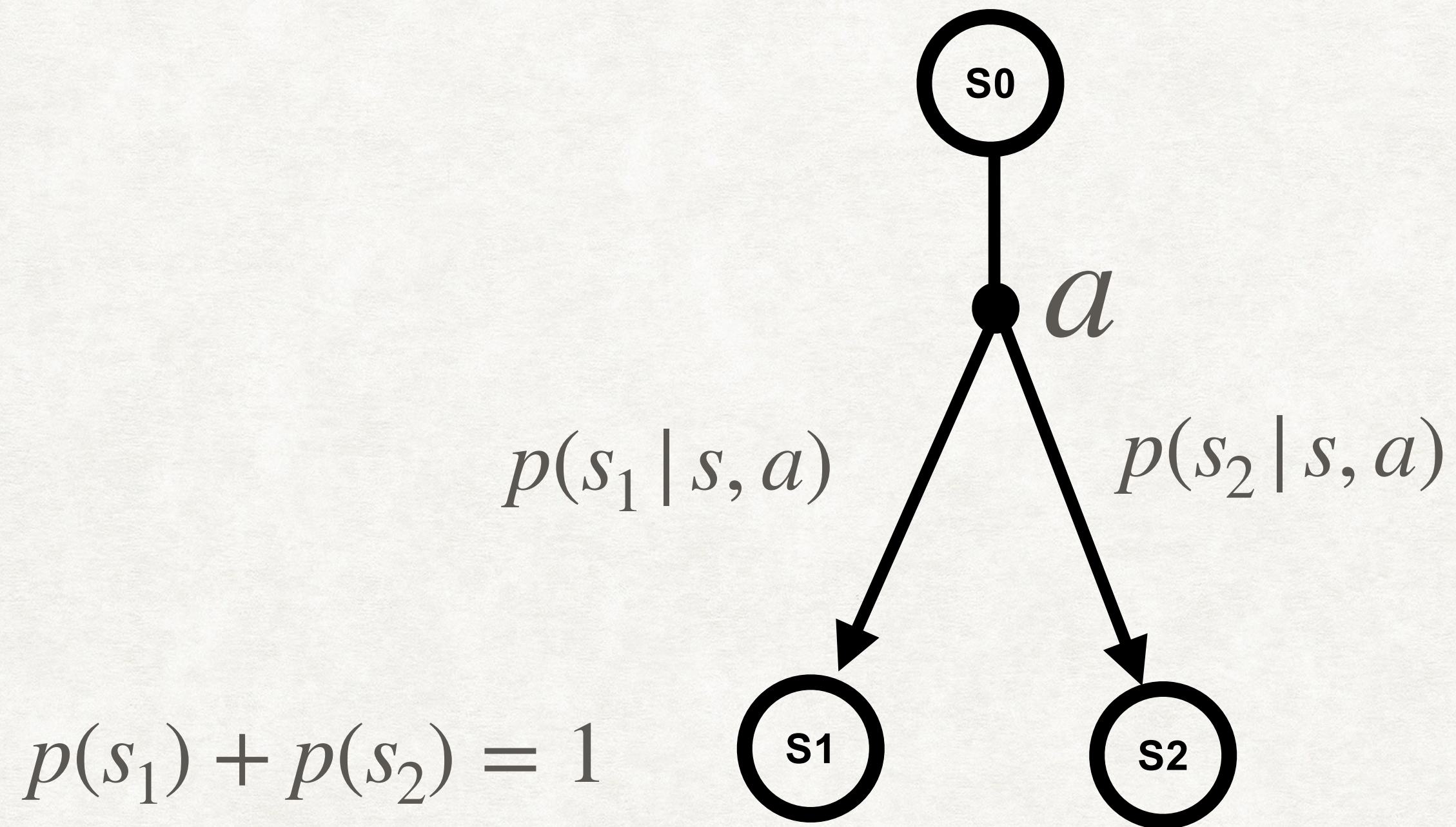
DETERMINISTIC EXAMPLE



Unless reward is explicitly mentioned assume $r=0$.

FUTURE REWARDS

STOCHASTIC EXAMPLE

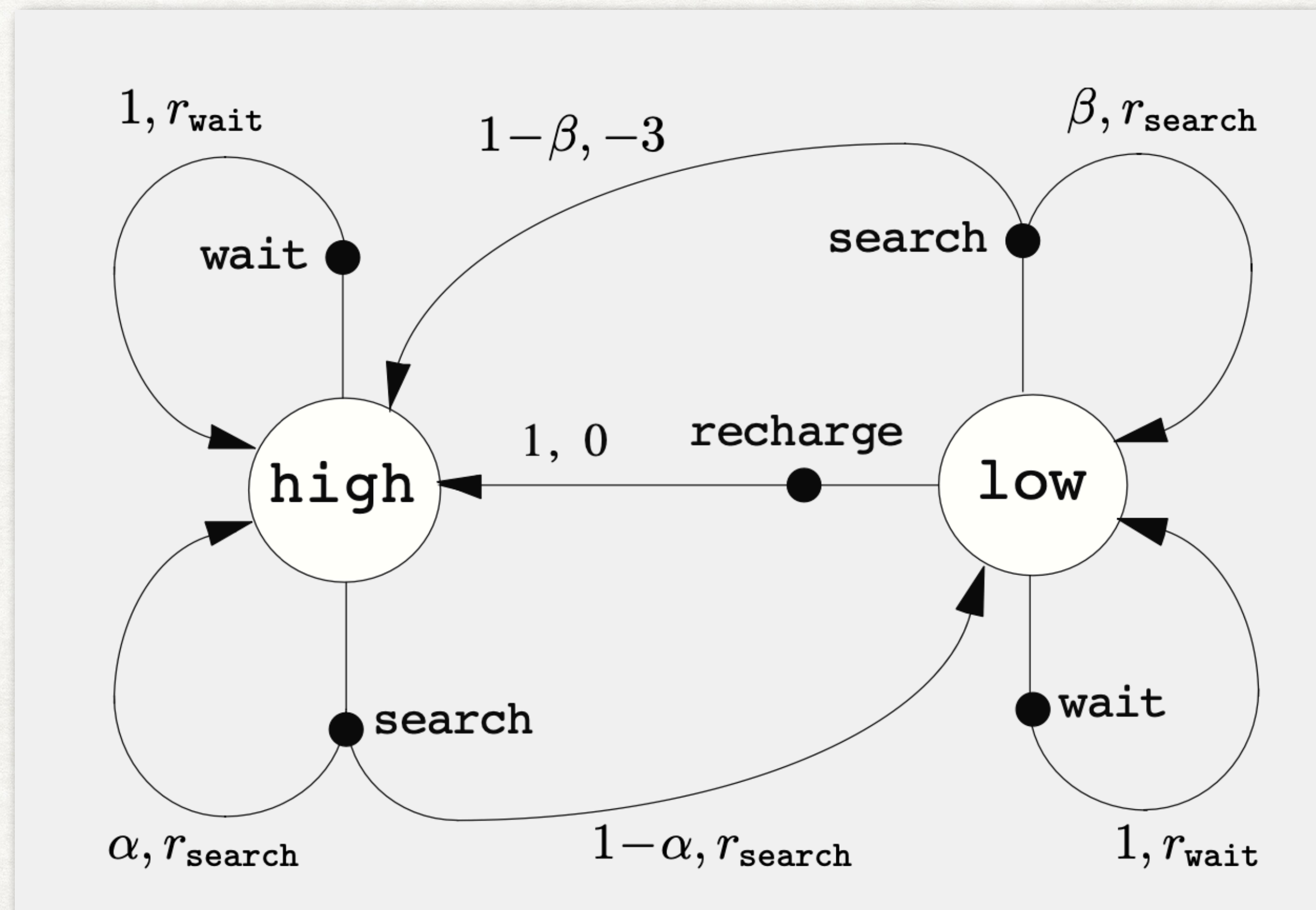


$$\sum_{s'} p(s' | S_t = s, A_t = a) = 1$$

$$s' \in \{s_1, s_2, \dots\}$$

FUTURE REWARDS

MARKOV DECISION PROCESS



Sutton & Barto, 2020 - with permission

FUTURE REWARDS

STATE AUGMENTATION



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FUTURE REWARD

EXPECTED RETURN

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

FUTURE REWARD

EXPECTED RETURN

$$0 \leq \gamma < 1$$



$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

FUTURE REWARDS

EXPECTED RETURN

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots)$$

$$G_{t+1} = R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots$$

$$G_t = R_{t+1} + \gamma G_{t+1}$$

FUTURE REWARDS

ACTION-VALUE FUNCTIONS

$$G_t = R_{t+1} + \gamma G_{t+1}$$

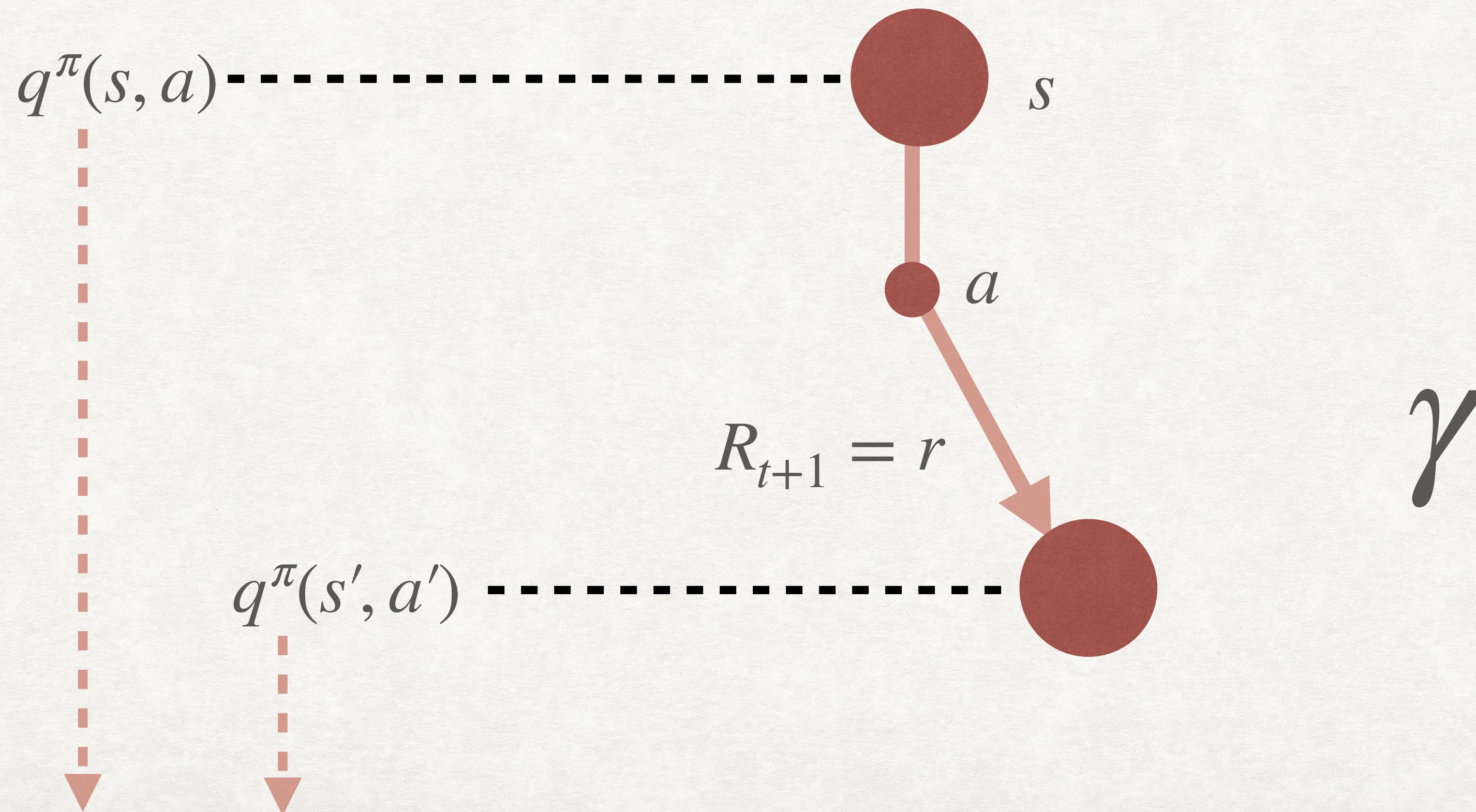
$$q^\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$

$$q^\pi(s, a) = \mathbb{E}_\pi[R_{t+1} + \gamma q^\pi(s', a') | S_t = s, A_t = a]$$

FUTURE REWARDS

ACTION-VALUE FUNCTIONS

$$q^{\pi}(s, a) = \mathbb{E}_{\pi}[R_{t+1} + \gamma q^{\pi}(s', a') \mid S_t = s, A_t = a]$$



FUTURE REWARDS

OPTIMALITY & BELLMAN EQUATION FOR ACTION-VALUES

$$q^{\pi}(s, a) = \mathbb{E}_{\pi}[R_{t+1} + \gamma q^{\pi}(s', a') \mid S_t = s, A_t = a]$$

$$q^*(s, a) = \max_{\pi} q^{\pi}(s, a)$$

$$q^*(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} q^*(s', a') \mid S_t = s, A_t = a]$$

FUTURE RETURNS

STATE VALUE FUNCTIONS

$$v(s) = \mathbb{E}[G_t | S_t = s]$$

$$v^*(s) = \max_a q^{\pi^*}(s, a)$$

SUMMARY

FUTURE REWARDS

- Expected returns and their recurrent relationship.
- Action-value functions q
- Optimal policy
- Optimal value function under optimal policy q^*
- “Greedy” selection of q -values - but here it is different from the bandit scenario, how?

THANK YOU!